Medical LLM Robustness

to noisy user inputs

We test whether prompt repair actually helps medical QA models answer messy real-user questions.

- Clean medical question
- Messy user question
- Repair rewrite
- Answer + evaluate

n=50 pilot | 11 models | fixed_repair + self_repair

Group 2 | Columbia University

01 Problem

Real users do not ask clean benchmark questions.

Medical QA systems see typos, vague wording, missing context, and patient-language phrasing.

A clean-input score tells us how a model behaves in the lab. It does not tell us what happens when a patient asks a messy question at midnight.

11
models

50
questions/model

5
noise types

Why this matters

In healthcare, a vague or misread question is not just a UX problem. It can change the medical concepts the model retrieves, the risks it mentions, and the confidence a user takes away.

Research question

Can a prompt-repair step recover answer quality lost under noisy user input?

02 Audience bridge

The project in one sentence

We take real medical questions, make them messy like real user input, try to repair them, then test whether LLM answers get better or worse.

Clean medical question > **Noisy user question** > **Repaired question** > **Model answer** > **Score vs reference**

Noise A controlled way of making the question harder: typo, vague phrasing, missing detail, layperson wording, or overgeneralization.

Repair A rewrite step that tries to make the messy question clearer without answering it.

Robustness Whether the model still gives a good answer when the user question is messy.

Recovery Whether repair moves the answer back toward the clean-question answer and reference.

03 Design

Two n=50 experiments separate fairness from realism.

The project now compares a controlled repair setup with a practical self-repair workflow.

Pipeline A	**Clean question -> answer model -> clean answer**
Pipeline B	**Noisy question -> answer model -> noisy answer**
Pipeline C	**Noisy question -> repair -> answer model -> repaired answer**

fixed_repair All 11 models see the same clean, noisy, and repaired questions. Best for apples-to-apples comparison.

self_repair All models see the same noisy questions, then each model repairs its own input. Best for practical workflow behavior.

04 Setup

What was tested

The completed n=50 pilot uses 11 answer models, five evenly assigned noise types, and 11 metrics per pipeline.

11
answer models
GPT-5.4 plus open-weight local/API models

50
questions/model
same pilot scale across both modes

5
noise types
10 examples each per model

550
rows/mode
11 models x 50 questions

Noise type	What it simulates
typos_grammar	spelling, punctuation, casing, grammar slips
ambiguity	blurred or underspecified medical detail
layperson	patient-style wording replacing technical terms
incomplete	missing condition, body part, timeframe, or qualifier
overgeneralization	specific query broadened into a generic health question

Model families	Examples
Closed/API	GPT-5.4
Open instruction	Llama 3.1, Mistral, Mixtral, Qwen, Gemma, Phi
Medical-domain	BioMistral-7B
Quantized local	4-bit and Q6_K variants

05 Evaluation

Metrics used in the project

The deck reports several metric families because no single metric captures medical QA robustness.

Family	Metric	What it measures
Lexical overlap	BLEU	word n-gram overlap with the reference answer
Lexical overlap	chrF	character n-gram F-score, more tolerant of wording/spelling shifts
Lexical overlap	ROUGE-L	longest common subsequence overlap
Lexical overlap	Token F1	normalized token precision/recall overlap
Lexical overlap	Exact Match	strict normalized full-answer match; mostly zero for free-form QA
Semantic answer	BERTScore	embedding-based similarity between answer and reference
Question intent	Intent Preservation	embedding similarity between clean question and noisy/repaired question
Judge score	G-Eval	GPT-5.4-mini score from 1-5 for answer quality
Medical content	med_coverage	medical-term recall vs the reference answer
Medical content	med_precision	medical-term precision in the model answer
Medical content	med_f1	balance of medical-term precision and coverage

06 Evaluation

How to read the metrics

A metric can improve because the question rewrite is cleaner, even if the final answer loses medical substance.

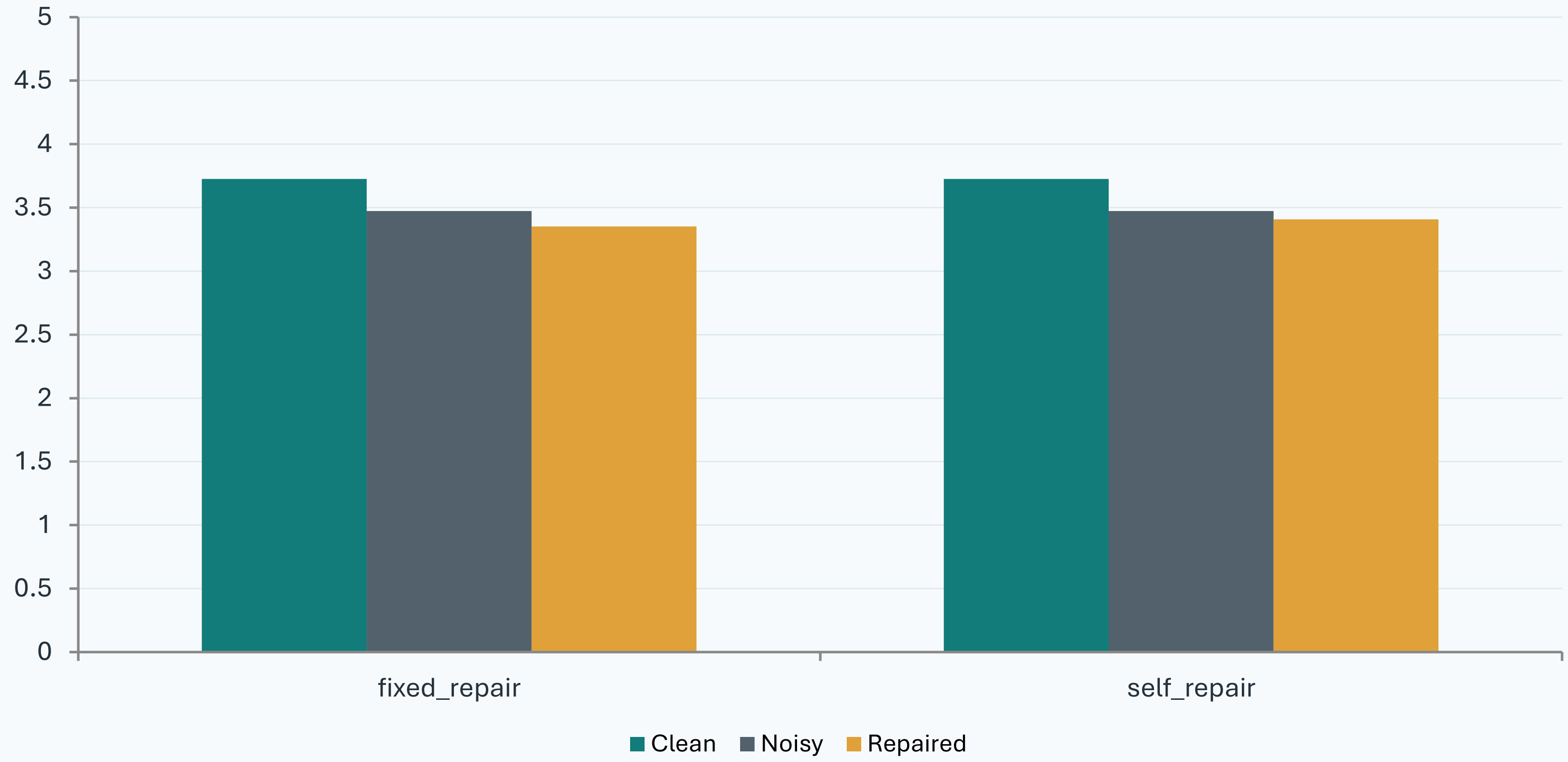
Lexical metrics	**similar words**	BLEU, chrF, ROUGE-L, Token F1, Exact Match
Semantic metrics	**similar meaning**	BERTScore and judge-style semantic similarity
Intent preservation	**same question intent**	Does the noisy or repaired question still point to the clean one?
Domain metrics	**medical content retained**	med_coverage, med_precision, med_f1 from a MedQuAD-derived lexicon
G-Eval	**judged answer quality**	GPT-5.4-mini sees question, reference answer, and model answer, then scores 1-5

Key reading rule: repair can clean up the question while still hurting the answer. That is why the deck pairs intent metrics with answer-quality and medical-content metrics.

07 Headline result

The n=50 pilot does not show systematic repair improvement.

Average G-Eval falls after noise; repair does not consistently recover it.



Mode averages

fixed_repair
3.73 → 3.47 → 3.35

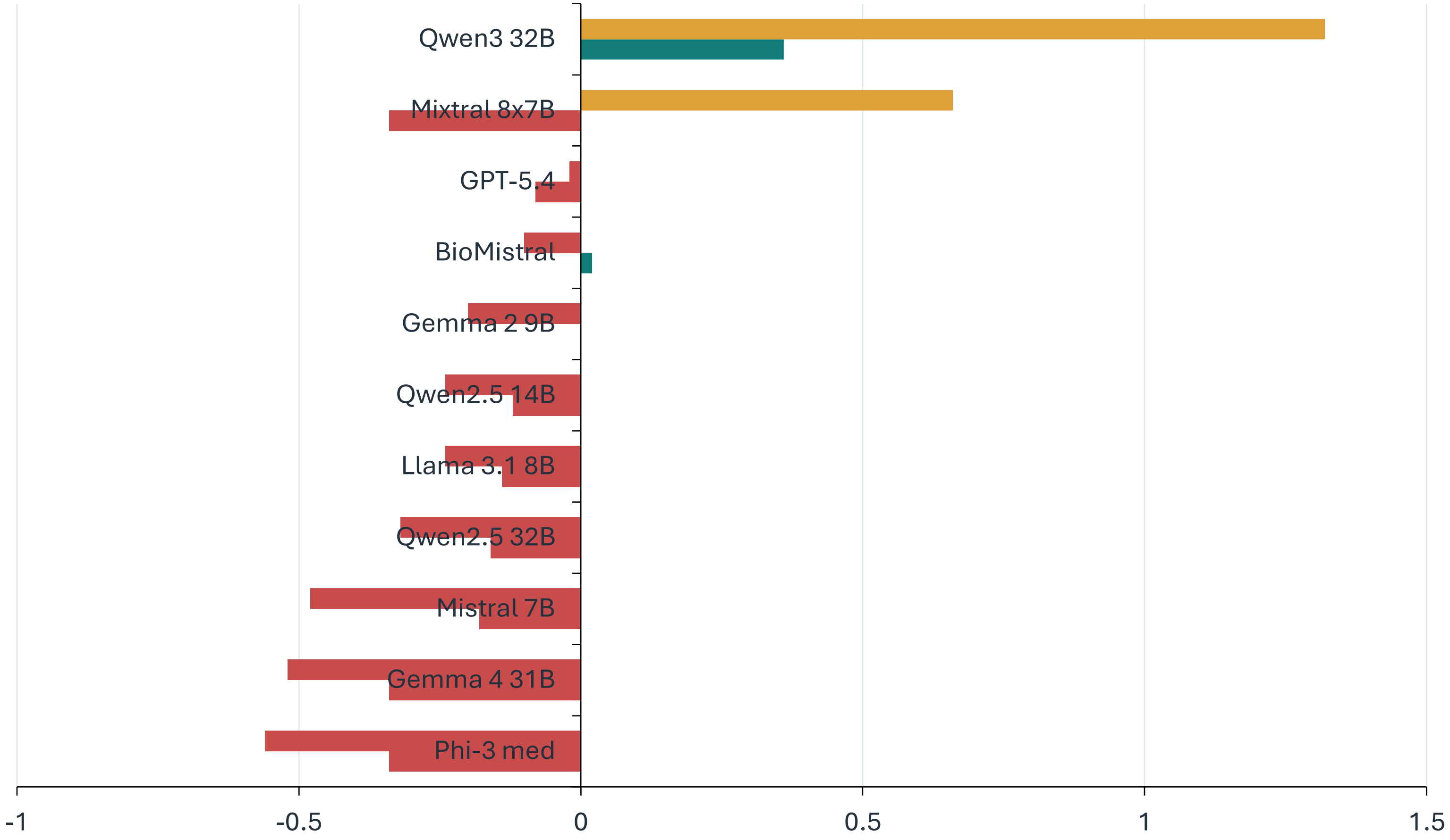
self_repair
3.73 → 3.47 → 3.41

Interpretation: repair is not a blanket safety layer. It sometimes helps, but the average n=50 answer-quality signal stays mixed.

08 Model comparison

Which models recover G-Eval after repair?

Repaired-minus-noisy G-Eval varies sharply by model and by repair mode.



What stands out

Self-repair has two large positive outliers: Qwen3 (+1.32) and Mixtral (+0.66).

Most other models are flat or worse after repair.

Fixed repair is more controlled but has smaller gains: only Qwen3 and BioMistral improve G-Eval.

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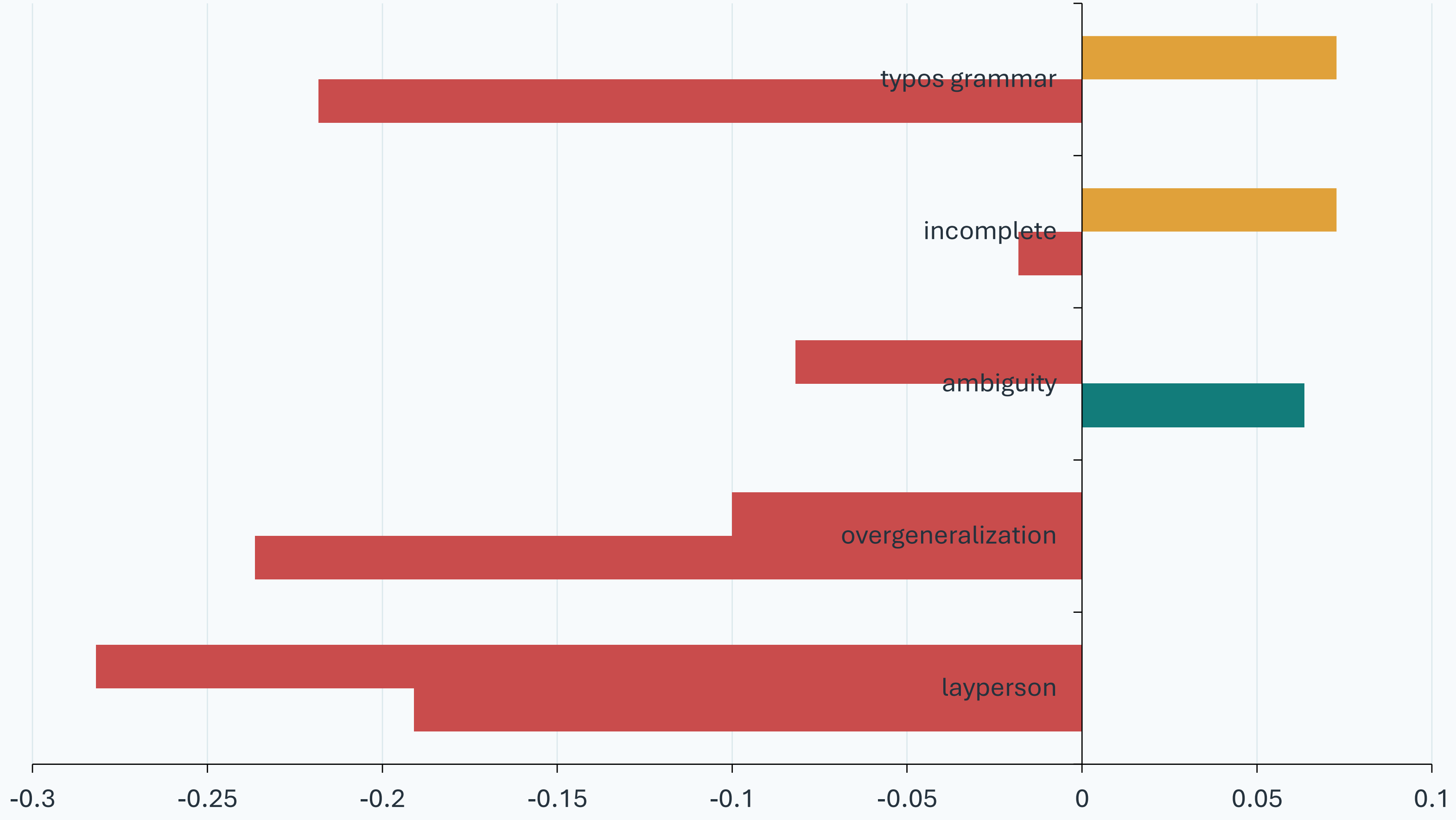
Noise breakdown

Repair behaves differently by noise type.

Bars show repaired-minus-noisy G-Eval. Values above zero mean repair helped answer quality.



Reading

Self-repair helps typos/grammar and incomplete questions slightly on average.

Layperson phrasing and overgeneralization remain difficult: repair often preserves or amplifies the wrong framing.

Averages hide examples where repair helps strongly and examples where it fails completely.

10 Metric tension

Repair cleans up intent more reliably than it improves answers.

Across 11 models, intent preservation improves everywhere, but answer and domain metrics often move the other way.

Metric	fixed: improved/worse/flat	self: improved/worse/flat
G-Eval	2 / 8 / 1	2 / 9 / 0
BERTScore	2 / 9 / 0	3 / 8 / 0
med_coverage	2 / 9 / 0	0 / 11 / 0
med_f1	4 / 7 / 0	1 / 10 / 0
Intent preservation	11 / 0 / 0	11 / 0 / 0

Core pattern **The repair step usually makes the question closer to the original intent. That does not guarantee the answer is better.**

11 Evidence example

Incomplete wording can make the answer collapse.

gpt-5.4 | incomplete noise | question_id 18

Question path

Do you have information about Air Pollution -> **do u have any info on bad air stuff and what it can do?** -> **What health effects can poor air quality or air pollution have on the body?**

CLEAN

Yes. Air pollution is the contamination of indoor or outdoor air by harmful substances that can affect health. Common air pollutants include: Particulate matter: PM2.5 and PM10 Ozone Nitrogen dioxide Sulfur dioxide Carbon monoxide Volatile organic compounds Smoke from vehicles, industry, wildfires, and tobacco Health effects can include: Eye, nose, and th...

NOISY

Yes - "bad air"

REPAIRED

Poor air quality can affect many parts of the body, especially the lungs and heart. Common health effects include: Eye, nose, and throat irritation Coughing, wheezing, shortness of breath Worsening of asthma, COPD, and allergies Reduced lung function and increased risk of respiratory infections like bronchitis More serious or long-term effects can include...

G-Eval: 4 clean | 1 noisy | 4 repaired

med_coverage: 27.1 -> 2.1 -> 16.7

The noisy question asks for 'bad air stuff' and GPT-5.4 answers with a fragment. Repair restores much of the answer quality.

12 Evidence example

Noisy wording can shift the topic.

gpt-5.4 | incomplete noise | question_id 28

Question path

Do you have information about Anabolic Steroids -> **do you hav any info on those muscle-building stuff?** -> **What are the health risks and side effects of using muscle-building supplements or anabolic steroids?**

CLEAN

Yes. Anabolic steroids usually refers to anabolic-androgenic steroids (AAS), which are synthetic versions of testosterone. What they do They can: Increase muscle mass and strength Promote protein synthesis Affect male sexual characteristics because of their androgenic effects Legitimate medical uses Doctors may prescribe them for certain conditions, such...

NOISY

Yes - if you mean muscle-building supplements or drugs, here's the short version: Things that actually help Protein powder: useful if you're not getting enough protein from food. Aim for about 1.6-2.2 g protein/kg body weight/day if trying to build muscle. Creatine monohydrate: one of the best-studied supplements for strength and muscle gain. Typical dose...

REPAIRED

Muscle-building products range from legal supplements to anabolic-androgenic steroids, and the risks depend heavily on what is being used. Muscle-building supplements Common products include protein powders, creatine, pre-workouts, "test boosters," and weight-gain products. Possible risks and side effects Stomach upset: nausea, bloating, diarrhea, cramps...

G-Eval: 4 clean | 2 noisy | 3 repaired

med_coverage: 28.3 -> 7.5 -> 13.2

The clean topic is anabolic steroids; the noisy answer drifts into generic supplements like protein powder and creatine.

13 Evidence example

Layperson phrasing changes the frame.

gpt-5.4 | layperson noise | question_id 2

Question path

What is (are) Abortion ? -> what does it mean when a pregancy gets ended early or the baby is not kept, like how does that whole thing happen and what is i... -> What does it mean when a pregnancy ends early or is intentionally ended, and what happens during that process?

CLEAN

Abortion is the termination or loss of a pregnancy before the fetus is viable outside the uterus. It may be: Spontaneous abortion: a miscarriage, occurring naturally. Induced abortion: deliberate ending of a pregnancy by medication or a medical/surgical procedure. In medical use, abortion generally refers to pregnancy ending in the early stages, before th...

NOISY

When a pregnancy ends early, there are a few different medical terms, depending on how it happens: Miscarriage This means the pregnancy ends on its own before the baby can survive outside the uterus, usually before 20 weeks. It often happens because the embryo/fetus was not developing normally, often due to chromosome problems. It is very common and usual...

REPAIRED

This is called abortion. If a pregnancy ends on its own early, it is a spontaneous abortion, also known as a miscarriage. If it is ended deliberately, it is an induced abortion. In either case, the pregnancy tissue-such as the embryo or fetus and the placenta-is expelled or removed from the uterus before the fetus is able to survive outside the womb. This...

G-Eval: 4 clean | 3 noisy | 4 repaired

med_coverage: 23.8 -> 28.6 -> 23.8

Here repair helps: it restores a concise medical distinction between spontaneous and induced abortion.

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Evidence example

A better judge score can still lose medical content.

Mixtral-8x7B-Instruct-v0.1 (4bit) | typos_grammar noise | question_id 30

Question path

What is (are) Anal Disorders ? -> what are anal disordrs exacly, and how do they happn ? -> What are anal disorders exactly, and how do they typically develop?

CLEAN

An anal disorder refers to various conditions that affect the anus, including: Hemorrhoids: Swollen veins in the rectum or anus that can cause pain, itching, and bleeding. Fissures: Small tears in the lining of the anus that can be painful and bleed during bowel movements. Abscesses: Infections that result in pus-filled pockets near the anus. Fistulas: Ch...

NOISY

Anal disorders refer to various conditions that affect the anus, including: Hemorrhoids: Swollen veins in the rectum or anus that can cause pain, itching, and bleeding. They can be caused by straining during bowel movements, obesity, pregnancy, standing or sitting for long periods, chronic constipation or diarrhea, eating a low-fiber diet, or aging. Fissu...

REPAIRED

Anal disorders refer to various conditions affecting the anus, including hemorrhoids, anal fissures, abscesses, and fistulas. They typically develop due to straining during bowel movements, constipation, diarrhea, aging, or anal intercourse. In some cases, they can be caused by underlying conditions like inflammatory bowel disease or colon cancer. Symptom...

G-Eval: 4 clean | 3 noisy | 5 repaired

med_coverage: 22.2 -> 19.4 -> 16.7

Mixtral self-repair improves G-Eval from 3 to 5, but med_coverage falls. This is why the metric stack matters.

15 Interpretation

What the n=50 evidence means

The project is not asking whether repair can produce nicer questions. It asks whether nicer questions produce better medical answers.

01 Repair is useful, but bounded.

It can fix surface noise and some missing-context cases, yet it cannot reliably recover medical meaning once the user prompt has drifted.

02 Question repair adds another failure mode.

The repair model can commit to the wrong interpretation or broaden the target before the answer model ever sees it.

03 Metric choice changes the story.

Intent scores often improve while G-Eval and medical-term coverage do not. The safest read uses multiple lenses.

16 **Limitations**

What this pilot cannot yet claim

The deck is deliberately conservative: these are completed n=50 pilot experiments, not the final 1,000-question study.

Limit	Why it matters
Pilot scale	50 questions per model. Good for signal-finding, not final statistical claim-making.
Synthetic noise	LLM-generated noise is controlled, but real patient language can be messier and multi-modal.
Single dataset	MedQuAD is curated NIH/NLM content, not patient portal messages or EHR notes.
Judge model overlap	G-Eval uses GPT-5.4-mini; the repair/noise stack also uses GPT-family components.
Domain metric limits	Medical-term coverage catches lost terms, but it does not judge clinical harm or relevance.

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Next steps

What would turn the pilot into a stronger study

The next phase should test scale, repair validation, and clinical safety rather than only adding more aggregate metrics.

Scale to n=1,000

Run the same fixed_repair and self_repair structure on a larger MedQuAD sample with confidence intervals.

Validate repair outputs

Reject rewrites that answer the question, ask for more context, or inject facts not present in the noisy input.

Human/clinical review

Pair automatic scores with a harm-tier rubric for misleading, unsafe, or clinically dangerous answers.

Noise-specific policy

Deploy repair selectively: typo/incomplete questions may benefit; overgeneralized or ambiguous prompts may need clarification instead.

Practical takeaway: prompt repair should be evaluated as a conditional tool, not assumed to be a universal safety layer.

Thank you.

Questions, pushback, and especially counter-evidence welcome.

Main answer

Repair improved question intent in every model, but it did not systematically improve medical QA answer quality in the completed n=50 pilot.

Group 2 | Practicum in LLM Evaluation

MedQuAD Robustness | n=50 final presentation